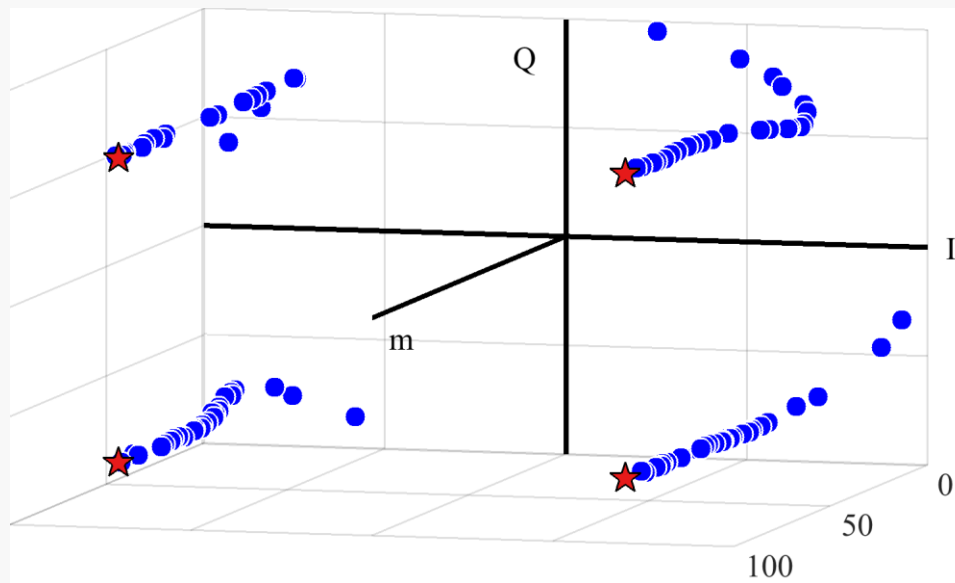


Carrier Phase Synchronization

Qasim Chaudhari



Introduction

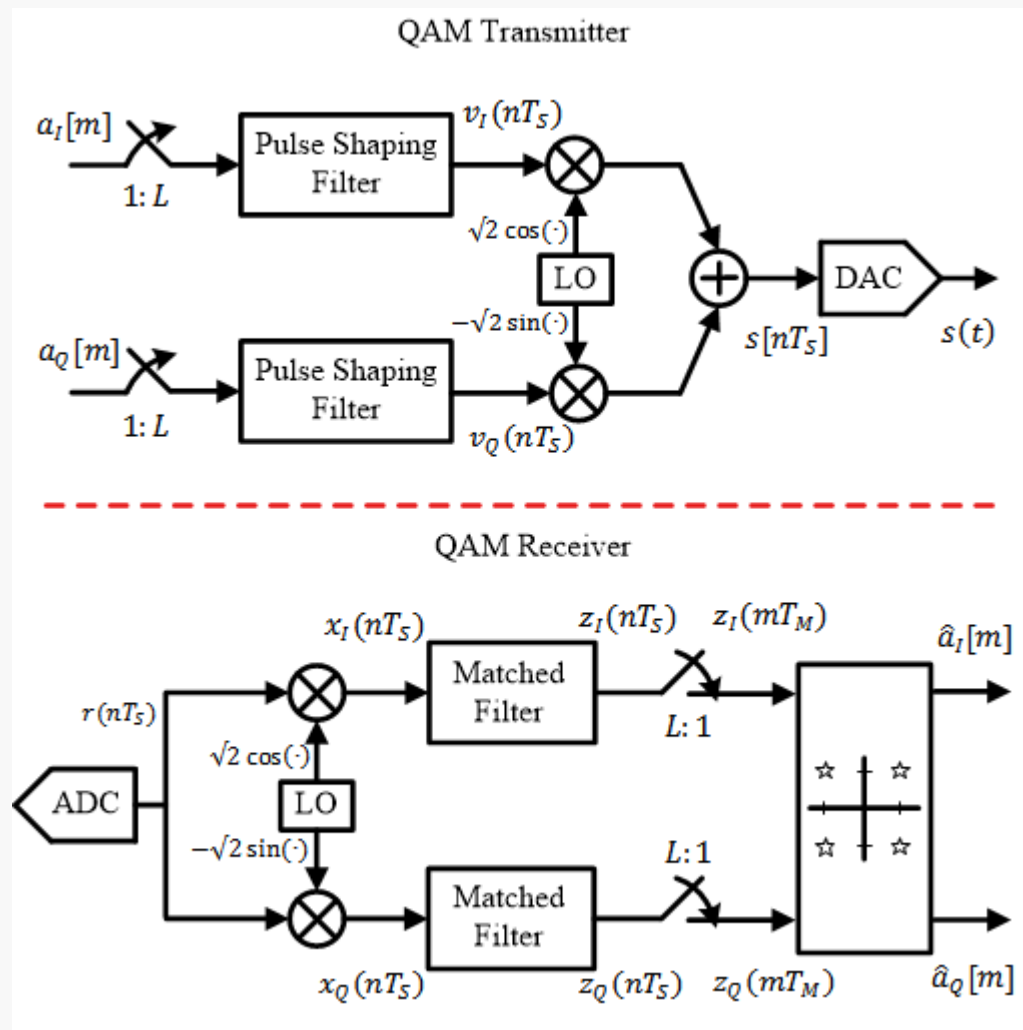
For QAM and PSK, the Rx LO is not phase and frequency aligned with the carrier of the incoming signal

A phase offset arises due to the following factors

- Tx time reference
- Rx time reference
- Overall channel impulse response
- Propagation delay from the Tx to the Rx

A QAM Transceiver

- Symbol rate $1/T_M$
- Rx sample rate $1/T_S$
- L samples/symbol, $T_S = T_M/L$
- Matched filter output $z(mT_M)$



Carrier Phase Mismatch

Matched filter output becomes

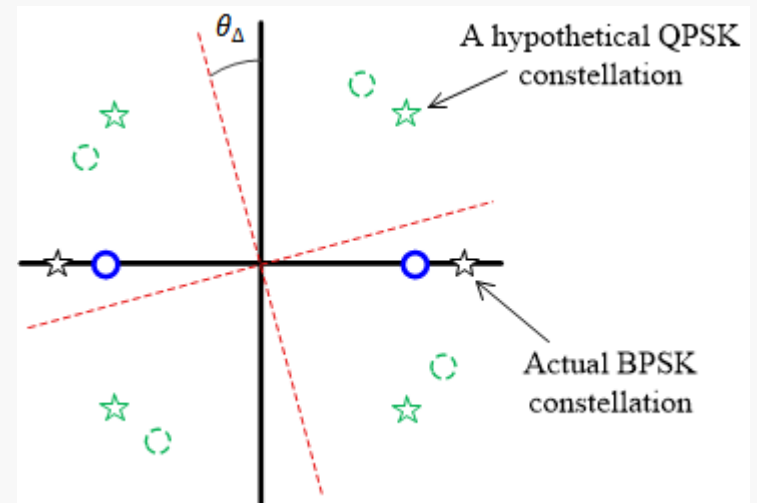
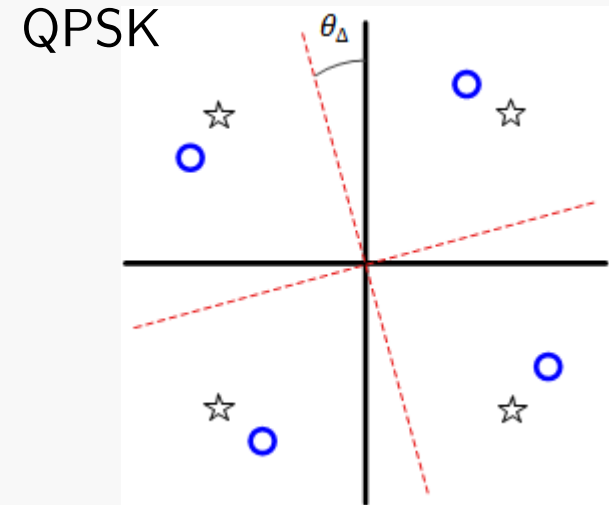
$$z_I(mT_M) = a_I[m] \cos \theta_\Delta - a_Q[m] \sin \theta_\Delta$$

$$z_Q(mT_M) = a_Q[m] \cos \theta_\Delta + a_I[m] \sin \theta_\Delta$$

In polar form,

$$|z(mT_M)| = \sqrt{a_I^2[m] + a_Q^2[m]}$$

$$\angle z(mT_M) = \angle a[m] + \theta_\Delta$$



Fundamental Problem of Synchronization

Signal at the Rx consists of two phase components

$$\angle z(mT_M) = \angle a[m] + \theta_\Delta$$

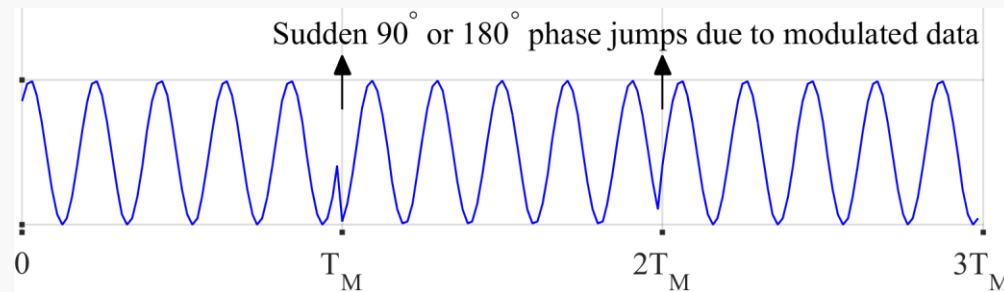
- From modulating data occurring at symbol rate
- Unknown phase difference θ_Δ between Tx and Rx local oscillator

Imagine a QPSK signal

$$s(t) = \sum_m \sqrt{a_I^2[m] + a_Q^2[m]} \cdot p(t - mT_M) \cdot \sqrt{2} \cos \left(2\pi F_C t + \tan^{-1} \frac{a_Q[m]}{a_I[m]} \right)$$

Fundamental Problem of Synchronization

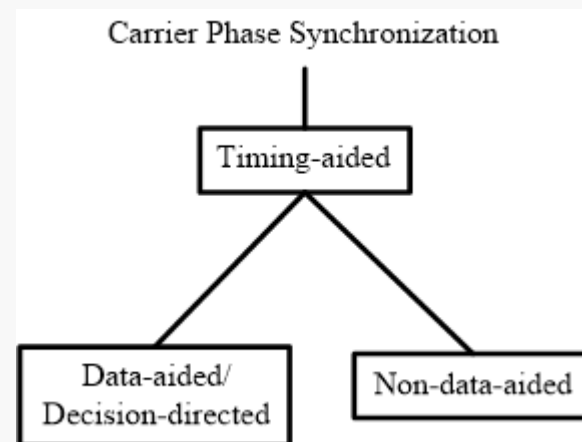
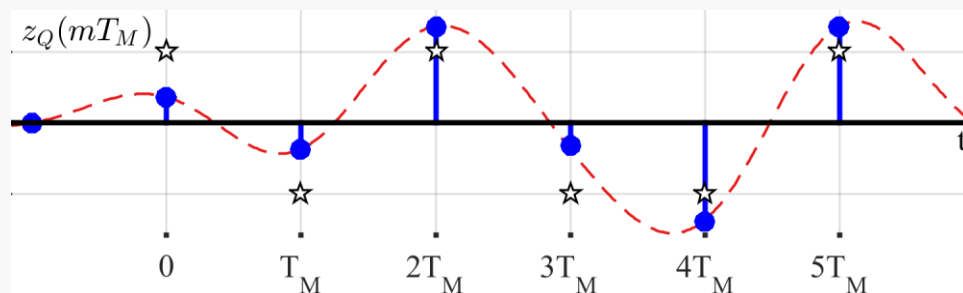
Signal phase is jumping around by 0° , $\pm 90^\circ$ or 180° at every symbol boundary due to the modulating data



This never allows a simple PLL to converge

The Big Picture

Almost all DSP based carrier phase synchronization algorithms are timing-aided



Enter the Correlation

- Correlation is the Master Algorithm for a digital communication system
- We apply the principle of maximum correlation for solving our phase synchronization problem
- Rx signal is correlated with a clean version of the expected signal template

Conjugate Product Estimator

From direct application of correlation

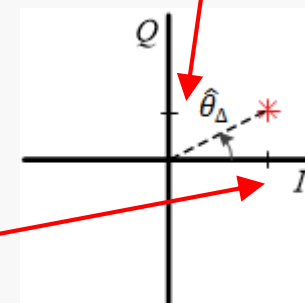
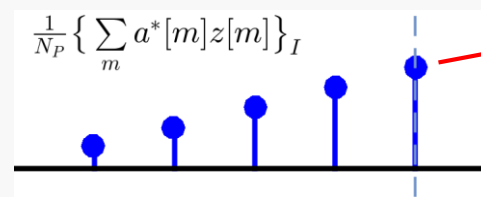
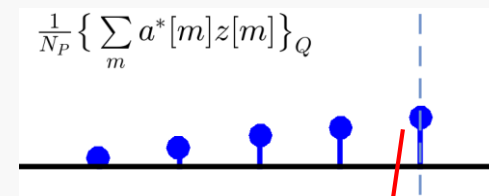
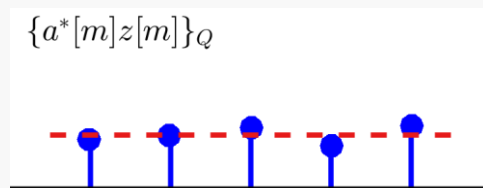
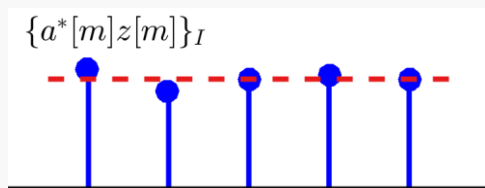
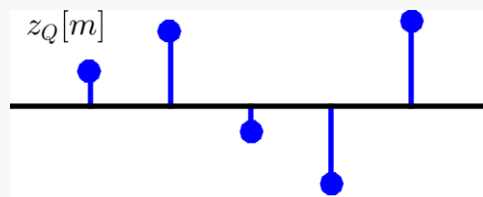
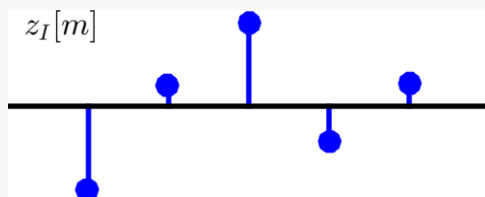
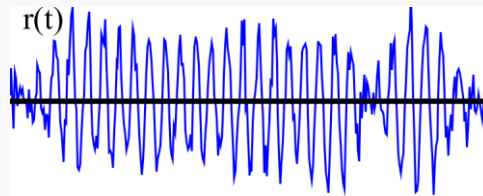
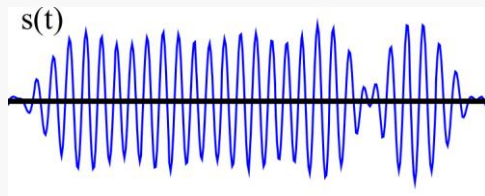
$$\hat{\theta}_{\Delta} = \angle \left\{ \sum_{m=0}^{N_0-1} a^*[m]z(mT_M) \right\}$$

- **Conjugate product:** operation $a^*[m]z(mT_M)$ removes the effect of phase shifts arising from data symbols

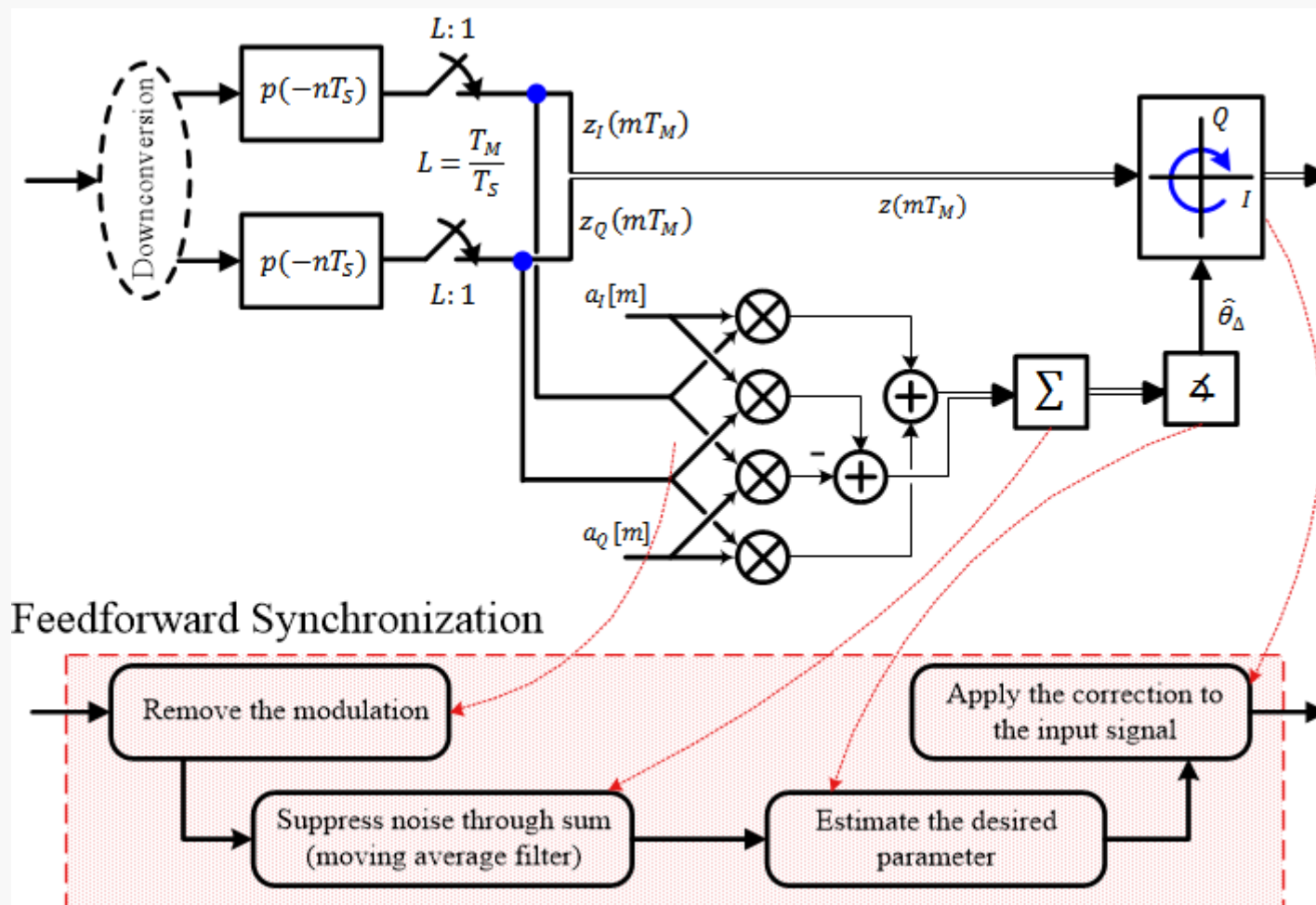
$$\angle \{a^*[m]z(mT_M)\} = \angle a^*[m] + \theta_{\Delta} - \angle a[m] = \theta_{\Delta}$$

- **Summation:** to reduce noise
- **Four-quadrant inverse tangent:** required to output $\hat{\theta}_{\Delta}$

Conjugate Product Estimator



Feedforward Synchronization



Phase Difference PED

PED = Phase Error Detector

Can a PLL be used for phase tracking?

Yes, if the PED can remove the modulation symbols

- At each symbol time mT_M , we de-rotate the matched filtered output $z(mT_M)$ by $\hat{\theta}_\Delta$ through an NCO

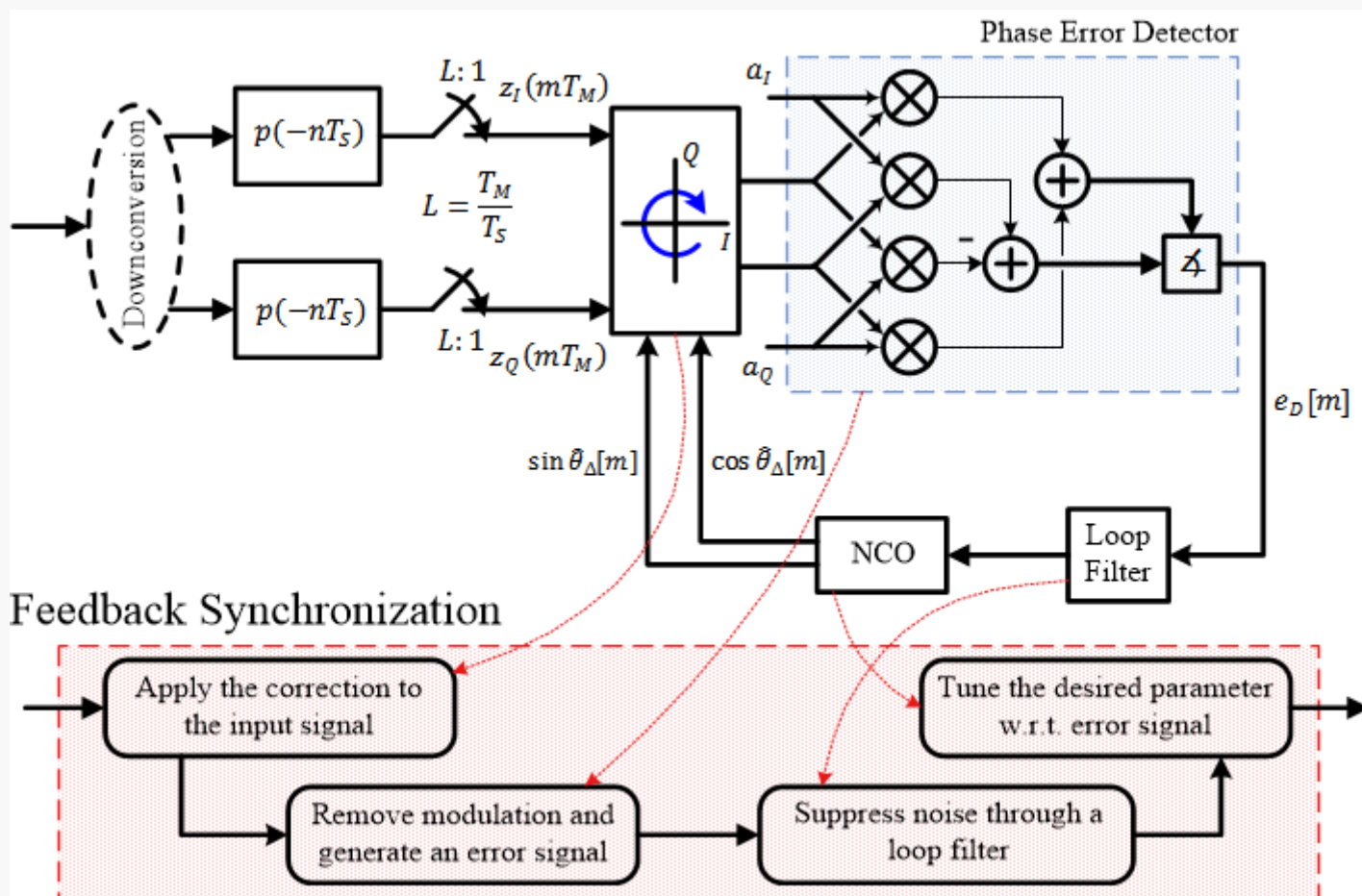
$$\hat{z}_I(mT_M) = z_I(mT_M) \cos \hat{\theta}_\Delta + z_Q(mT_M) \sin \hat{\theta}_\Delta$$

$$\hat{z}_Q(mT_M) = z_Q(mT_M) \cos \hat{\theta}_\Delta - z_I(mT_M) \sin \hat{\theta}_\Delta$$

- Use the phase of the conjugate product as a phase error detector

$$e_D[m] = \angle \left\{ a^*[m] \hat{z}(mT_M) \right\}$$

Feedback Synchronization



Cross Product PED

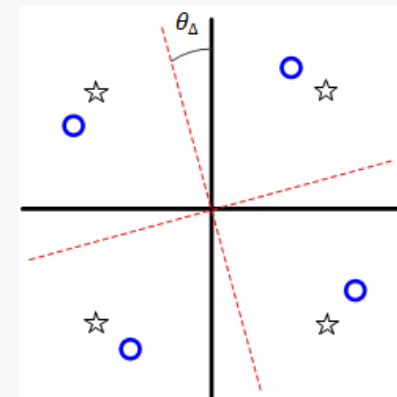
Write our old $\hat{\theta}_\Delta = \angle \left\{ \sum_{m=0}^{N_0-1} a^*[m]z(mT_M) \right\}$ in IQ form

Above expression is given by

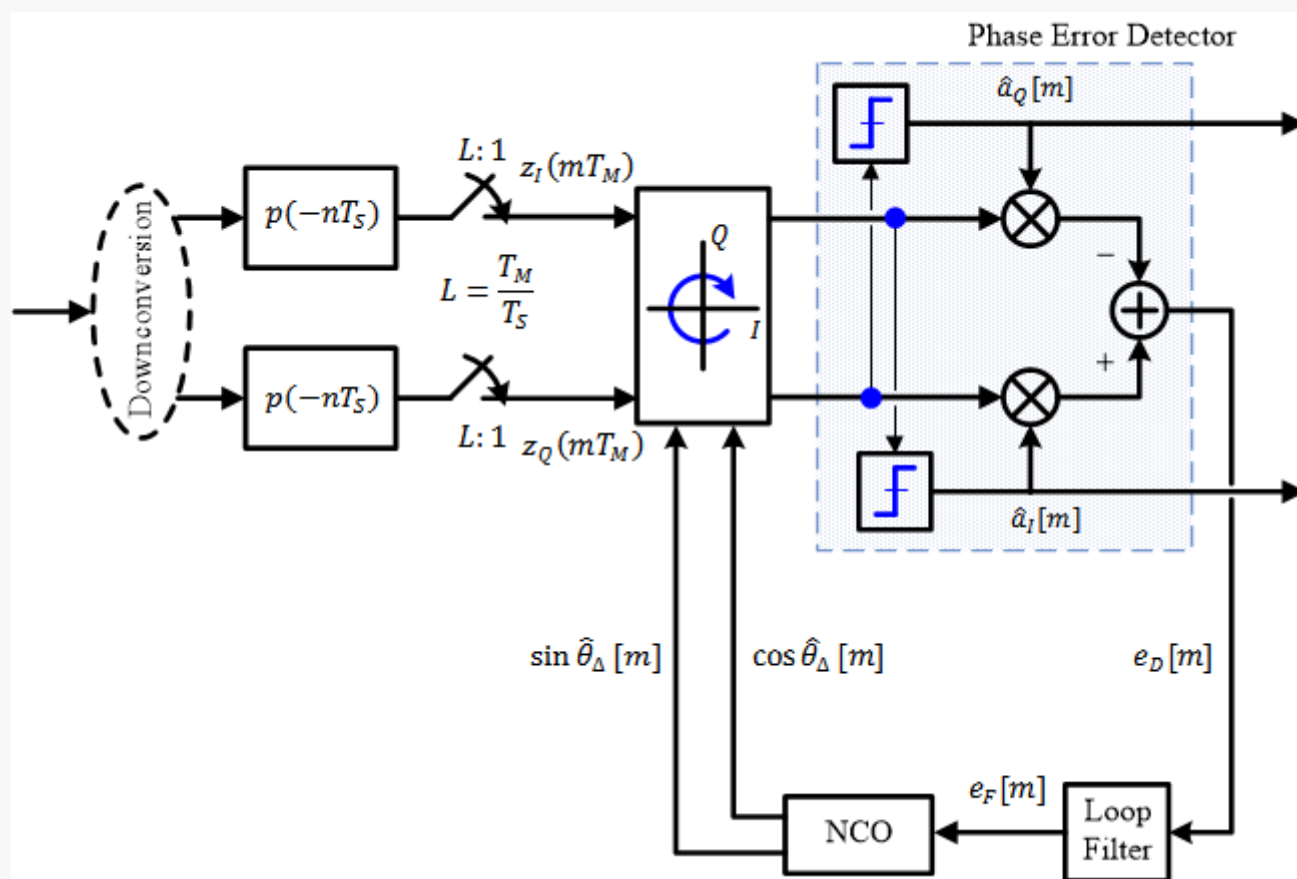
$$\begin{aligned} I &\rightarrow a_I[m]z_I(mT_M) + a_Q[m]z_Q(mT_M) \\ Q &\rightarrow a_I[m]z_Q(mT_M) - a_Q[m]z_I(mT_M) \end{aligned}$$

Main idea: Q term approaches zero, nullifying the effect of θ_Δ

$$e_D[m] = a_I[m]\hat{z}_Q(mT_M) - a_Q[m]\hat{z}_I(mT_M)$$



Cross Product PED



Example

Phase recovery process of a modulated signal through a PLL

$$B_n T_M = 0.05, \quad \zeta = 0.8$$

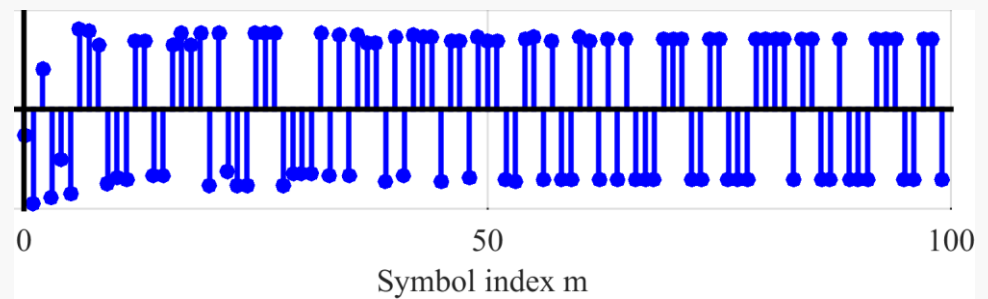
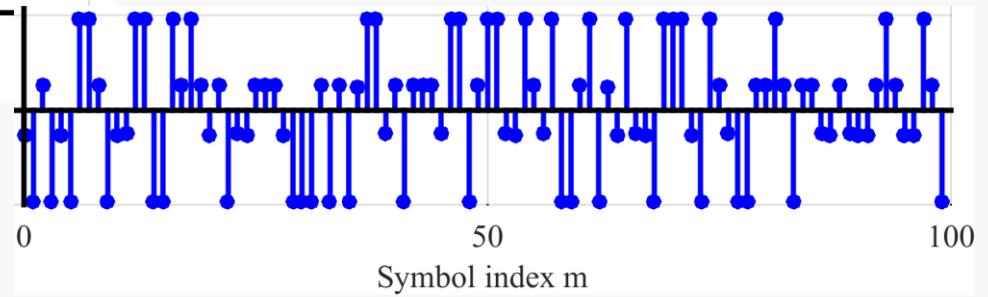
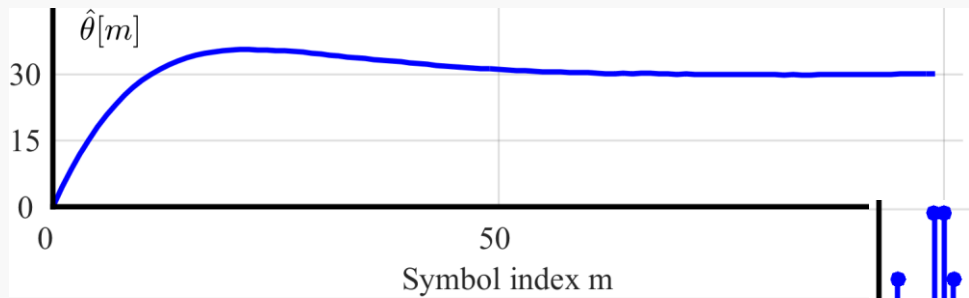
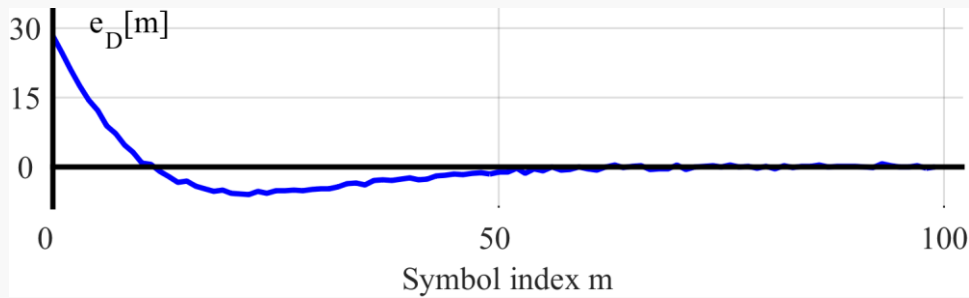
For a unit energy QPSK modulation $A = 1/\sqrt{2}$

PI loop filter coefficients are given by

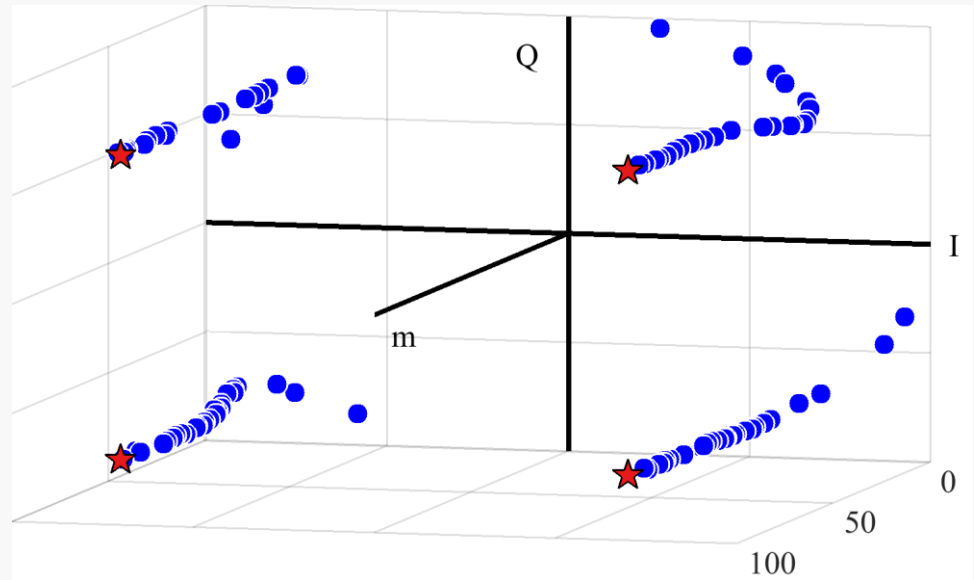
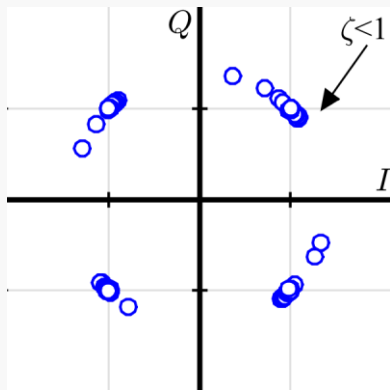
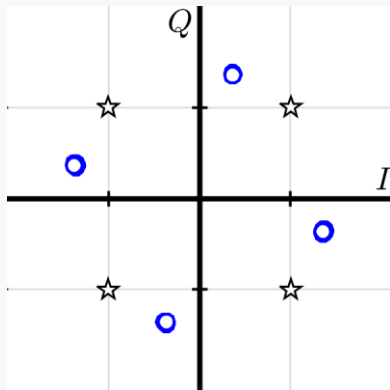
$$K_P = 0.1438$$

$$K_i = 0.0081$$

Example



Example



Mth Power Estimator

Raising to fourth power maps QPSK constellation points all to the same point (thus removing the modulation)

$$4 \cdot \left\{ \pm \frac{\pi}{4}, \pm \frac{3\pi}{4} \right\} = \pi$$

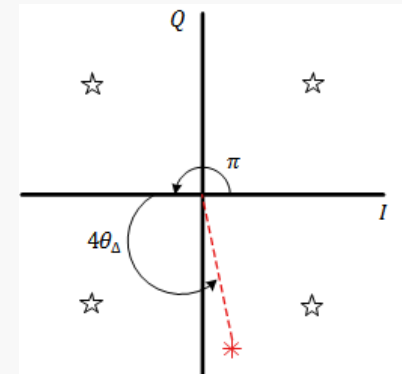
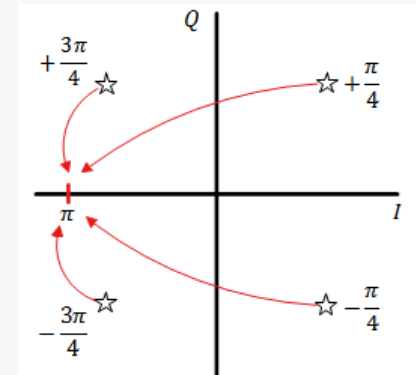
In the presence of θ_Δ

$$\angle z(mT_M) = \angle a[m] + \theta_\Delta$$

$$\angle z^4(mT_M) = 4 \cdot \angle a[m] + 4\theta_\Delta = \pi + 4\theta_\Delta$$

Estimator expression

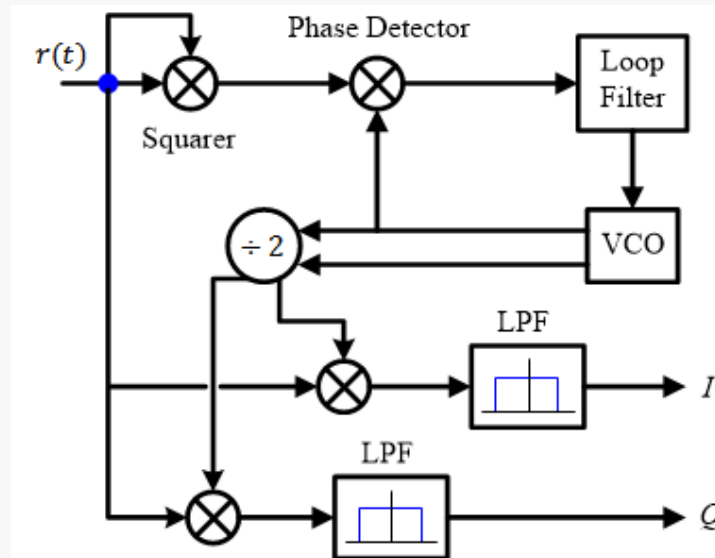
$$\hat{\theta}_\Delta = \frac{1}{4} \cdot \angle \left\{ \sum_{m=0}^{N_0-1} z^4(mT_M) \right\} - \frac{\pi}{4}$$



Squaring Loop

Feedback solution is also possible

Special case of an M^{th} power phase error detector for $M = 2$, (BPSK modulation)



Costas Loop: BPSK Modulation

To understand its operation

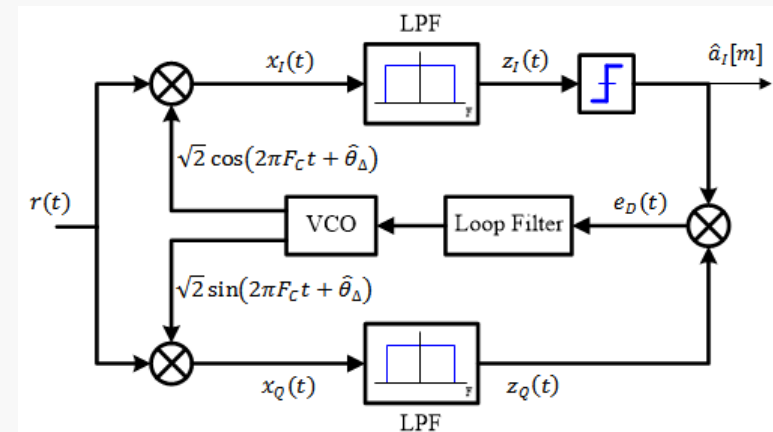
$$z_I(t) = v_I(t) \cos \theta_{\Delta:e}$$

$$z_Q(t) = -v_I(t) \sin \theta_{\Delta:e}$$



Error signal is generated as

$$\begin{aligned} e_D(t) &= -v_I^2(t) \cos \theta_{\Delta:e} \sin \theta_{\Delta:e} \\ &= -\frac{1}{2} v_I^2(t) \sin 2\theta_{\Delta:e} \end{aligned}$$



Costas Loop: QPSK Modulation

- Interesting principle of operation

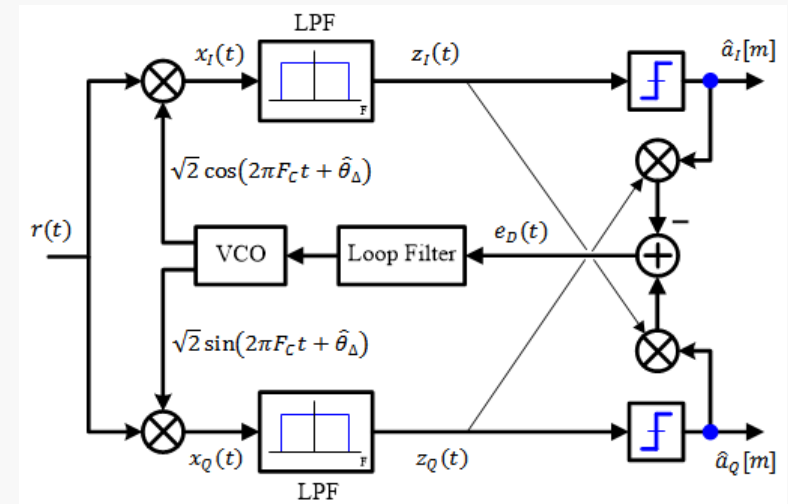
$$\text{sign}\{z_I(t)\} \approx \hat{a}_I[m], \quad \text{sign}\{z_Q(t)\} \approx \hat{a}_Q[m]$$

- Error signal is generated as

$$\begin{aligned} e_D(t) &= \text{sign}\{z_Q(t)\}z_I(t) - \text{sign}\{z_I(t)\}z_Q(t) \\ &\approx \hat{a}_Q[m]z_I(t) - \hat{a}_I[m]z_Q(t) \end{aligned}$$

- Compare with cross product PED

$$e_D[m] = \hat{a}_I[m]\hat{z}_Q(mT_M) - \hat{a}_Q[m]\hat{z}_I(mT_M)$$

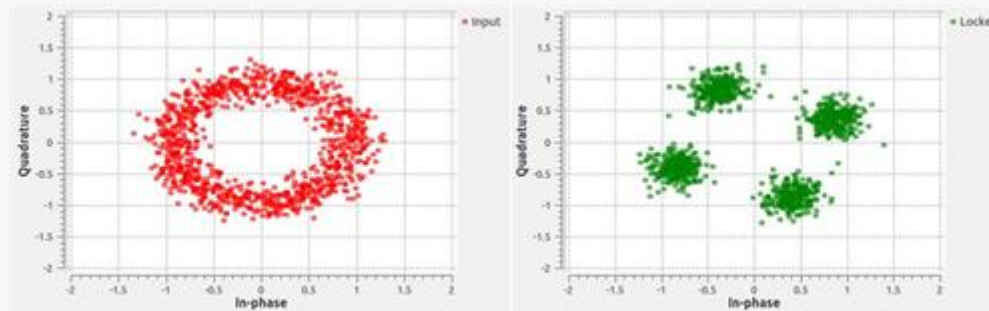


GNU Radio

Costal Loop block

- Loop bandwidth: equivalent noise bandwidth covered in PLL
- Order: modulation order (2 for BPSK, 4 for QPSK)
- Use SNR: more accurate phase estimation

Example: loop bandwidth $2\pi/100$, QPSK modulation and CFO -0.043



Small Picture

- Matched filtered signal is downsampled to 1 sample/symbol
- Modulation is removed through data symbol $a^*[m]$
- Phase of the resultant complex number